Name: Hashir Ahmed Buriro **Python Programming Internship** **Student ID: CA/SE3/12742**

Task 1: Hangman Game

Goal: Create a simple text-based Hangman game where the player guesses a word one letter at a time.

Simplified Scope:

- Use a small list of 5 predefined words (no need to use a file or API).
- Limit incorrect guesses to 6.
- Basic console input/output no graphics or audio.

Key Concepts Used: random, while loop, if-else, strings, lists.

CODE:

```python
import random

# List of 5 predefined words
words = ["apple", "chair", "tiger", "plant", "bread"]

# Randomly select a word
word = random.choice(words)

# Initialize variables
guessed_letters = []
incorrect_guesses = 0
max_attempts = 6

print("🎮 Welcome to Hangman Game!")

# Game loop
while incorrect_guesses < max_attempts:

    # Display current progress
    display_word = ""
    for letter in word:
        if letter in guessed_letters:
            display_word += letter + " "
        else:
            display_word += "_ "

    print("\nWord:", display_word)

    # Check win condition
    if "_" not in display_word:
        print("🎉 Congratulations! You guessed the word:", word)
        break

    # Take input
    guess = input("Enter a letter: ").strip().lower()

    # Validate input
    if len(guess) != 1 or not guess.isalpha():
        print("⚠️ Please enter a single valid letter!")
        continue

    # Check if already guessed
    if guess in guessed_letters:
        print("⚠️ You already guessed that letter!")
```

```python
    if len(guess) != 1 or not guess.isalpha():
        print("⚠️ Please enter a single valid letter!")
        continue

    # Check if already guessed
    if guess in guessed_letters:
        print("⚠️ You already guessed that letter!")
        continue

    # Add guess to list
    guessed_letters.append(guess)

    # Check correct or wrong
    if guess in word:
        print("✅ Correct!")
    else:
        incorrect_guesses += 1
        print("❌ Wrong! Attempts left:", max_attempts - incorrect_guesses)

# Lose condition
if incorrect_guesses == max_attempts:
    print("💀 Game Over! The word was:", word)
```

OUTPUT:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Office Remote Work\Code Alpha Remote Internship 3 months\Tasks>  & 'c:\Users\burir\AppData\
2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '62374' '--' 'd:\Office Remote Work\Code Alpha
🎮 Welcome to Hangman Game!

Word: _ _ _ _ _
Enter a letter: a
✅ Correct!

Word: _ _ a _ _
Enter a letter: s
❌ Wrong! Attempts left: 5

Word: _ _ a _ _
Enter a letter: c
✅ Correct!

Word: c _ a _ _
Enter a letter: h
✅ Correct!

Word: c h a _ _
Enter a letter: r
✅ Correct!

Word: c h a _ r
Enter a letter: i
✅ Correct!

Word: c h a i r
🎉 Congratulations! You guessed the word: chair
PS D:\Office Remote Work\Code Alpha Remote Internship 3 months\Tasks>
```

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```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Office Remote Work\Code Alpha Remote Internship 3 months\Tasks>  d:; cd 'd:\Office Remote Work\Code Alpha Remote Internship 3
re-3.14-64\python.exe' 'c:\Users\burir\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '5985
months\Tasks\Hashir Ahmed Task 1 The Hangmen Game.py'
Word: _ h _ _ _
Enter a letter: i
✅ Correct!

Word: _ h _ i _
Enter a letter: i
⚠️You already guessed that letter!

Word: _ h _ i _
Enter a letter: i
⚠️You already guessed that letter!

Word: _ h _ i _
Enter a letter: d
❌ Wrong! Attempts left: 5

Word: _ h _ i _
Enter a letter: w
❌ Wrong! Attempts left: 4

Word: _ h _ i _
Enter a letter: w
⚠️You already guessed that letter!

Word: _ h _ i _
Enter a letter: e
❌ Wrong! Attempts left: 3

Word: _ h _ i _
Enter a letter: x
❌ Wrong! Attempts left: 2

Word: _ h _ i _
Enter a letter: v
❌ Wrong! Attempts left: 1

Word: _ h _ i _
Enter a letter: f
❌ Wrong! Attempts left: 0
💀 Game Over! The word was: chair
PS D:\Office Remote Work\Code Alpha Remote Internship 3 months\Tasks>
                                                                    Start JSON S
```

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TASK 2: Stock Portfolio Tracker

- **Goal:** Build a simple stock tracker that calculates total investment based on manually defined stock prices.
- **Simplified Scope:**
  - User inputs stock names and quantity.
  - Use a hardcoded dictionary to define stock prices (e.g., {"AAPL": 180, "TSLA": 250}).
  - Display total investment value and optionally save the result in a .txt or .csv file.
- **Key Concepts Used:** dictionary, input/output, basic arithmetic, file handling (optional)

CODE:

Hashir Ahmed Task 2 Stock Portfolio Tracker.py

```python
# TASK 2: Stock Portfolio Tracker

# Step 1: Hardcoded stock prices
stock_prices = {
    "AAPL": 180,
    "TSLA": 250,
    "GOOGL": 2800,
    "AMZN": 3300,
    "MSFT": 300
}

# Step 2: Initialize portfolio dictionary
portfolio = {}

print("📈 Welcome to Stock Portfolio Tracker!")
print("Available stocks and their prices:")
for stock, price in stock_prices.items():
    print(f"{stock}: ${price}")

# Step 3: User input
while True:
    stock_name = input("\nEnter stock symbol (or 'done' to finish): ").upper()
    if stock_name == "DONE":
        break
    if stock_name not in stock_prices:
        print("⚠️ Stock not available. Try again.")
        continue
    try:
        quantity = int(input(f"Enter quantity of {stock_name}: "))
        if quantity < 0:
            print("⚠️ Quantity cannot be negative.")
            continue
    except ValueError:
        print("⚠️ Enter a valid number.")
        continue

    # Save in portfolio
    if stock_name in portfolio:
        portfolio[stock_name] += quantity
    else:
        portfolio[stock_name] = quantity

# Step 4: Calculate total investment
total_investment = 0
print("\n✅ Your Portfolio Summary:")
for stock, qty in portfolio.items():
    price = stock_prices[stock]
    investment = price * qty
```

```python
        portfolio[stock_name] = quantity

# Step 4: Calculate total investment
total_investment = 0
print("\n✅ Your Portfolio Summary:")
for stock, qty in portfolio.items():
    price = stock_prices[stock]
    investment = price * qty
    total_investment += investment
    print(f"{stock}: {qty} shares x ${price} = ${investment}")

print(f"\nTotal Investment Value: ${total_investment}")

# Step 5: Optional - Save to a text file
save_option = input("\nDo you want to save this summary to a file? (yes/no): ").lower()
if save_option == "yes":
    with open("portfolio_summary.txt", "w") as f:
        f.write("Stock Portfolio Summary\n")
        f.write("====================\n")
        for stock, qty in portfolio.items():
            price = stock_prices[stock]
            investment = price * qty
            f.write(f"{stock}: {qty} shares x ${price} = ${investment}\n")
        f.write(f"\nTotal Investment Value: ${total_investment}\n")
    print("✅ Portfolio summary saved as portfolio_summary.txt")
```

OUTPUT:

```
PS D:\Office Remote Work\Code Alpha Remote Internship 3 months\Tasks>  & 'c:\Users\burir\AppData\Lo
sions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '59914' '--' 'd:\Office
AMZN: $3300
MSFT: $300

Enter stock symbol (or 'done' to finish): AAPL
Enter quantity of AAPL: 52000

Enter stock symbol (or 'done' to finish): appl
⚠️Stock not available. Try again.

Enter stock symbol (or 'done' to finish): XXD
⚠️Stock not available. Try again.

Enter stock symbol (or 'done' to finish): TSLA
Enter quantity of TSLA: 2250

Enter stock symbol (or 'done' to finish): GOOGL
Enter quantity of GOOGL: 55566

Enter stock symbol (or 'done' to finish): AMZN
Enter quantity of AMZN: 555

Enter stock symbol (or 'done' to finish): MSFT
Enter quantity of MSFT: 999

Enter stock symbol (or 'done' to finish): akk
⚠️Stock not available. Try again.

Enter stock symbol (or 'done' to finish): done

✅ Your Portfolio Summary:
AAPL: 52000 shares x $180 = $9360000
TSLA: 2250 shares x $250 = $562500
GOOGL: 55566 shares x $2800 = $155584800
AMZN: 555 shares x $3300 = $1831500
MSFT: 999 shares x $300 = $299700

Total Investment Value: $167638500

Do you want to save this summary to a file? (yes/no): yes
✅ Portfolio summary saved as portfolio_summary.txt
PS D:\Office Remote Work\Code Alpha Remote Internship 3 months\Tasks>
```